

# NAMAN MANOCHA

## AI ENGINEER — Applied ML, RAG & Production AI Systems

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### SUMMARY

AI/ML engineer who ships end-to-end research products spanning data pipelines, retrieval and model logic, backend APIs, and deployment. Independently developed FundersAI, a research platform combining deterministic financial metrics, hybrid retrieval, and explainable ML clustering, and CareFlow Intelligence, an agentic RAG and fine-tuning system. Currently developing AI products independently. Core technologies: Python, FastAPI, Next.js, Supabase/PostgreSQL, Docker, and Kubernetes.

### TECHNICAL SKILLS

ML & AI: LangGraph, RAG, Hybrid Search, pgvector, ChromaDB, K-Means Clustering, Logistic Regression, MLflow, scikit-learn, Prompt Engineering, Hugging Face, Langfuse

Backend & Cloud: Python, FastAPI, REST APIs, SQL, PostgreSQL, Supabase, Redis, Docker, Kubernetes, GitHub Actions, Google Cloud Run

Web & Data: Next.js, React, TypeScript, ETL Pipelines, Web Scraping, Data Validation

### EXPERIENCE

#### Technology Manager — PayGain Multiservices Pvt. Ltd.

Feb 2025 – Present

Kolkata, India

- Scoping and leading development of a company-wide mobile app as sole engineer — client/employee onboarding, document access, product activation, and payments.
- Designed and deployed a lead-capture automation (Make) triggered on Google Form submissions; also drove 30+ client conversions by building and managing the company website and digital discovery flows.

### INDEPENDENT PROJECTS

#### FundersAI — AI-Powered Research Platform for Indian Stocks & Mutual Funds

[fundersai.co.in](https://fundersai.co.in)

Python, FastAPI, Next.js, Supabase/pgvector, LangGraph, Groq, OpenAI, Cloudflare R2 | ~141K source LOC, 841 files, 441 commits, 170 test files — OpenAI Build Week | Live: [www.fundersai.co.in](https://www.fundersai.co.in) | Repo: [github.com/Naman6019/FundersAI](https://github.com/Naman6019/FundersAI)

- Citation-Grounded RAG: built a seven-stage LangGraph workflow with retrieval grading, query rewriting, citation-aware synthesis, and explicit abstention when evidence is insufficient.
- Explainable ML:** fund-similarity engine using cosine similarity and k-means clustering over 13 standardized features, plus a logistic-regression triage model (MLflow) for parser-review prioritization.
- Data Engineering: automated collection and normalization of official mutual-fund disclosures, with data-quality checks and targeted normalizer fixes for affected records.

#### CareFlow Intelligence — Agentic RAG & Fine-Tuning System

Python, FastAPI, Next.js, Supabase/pgvector, ChromaDB, Unsloth, Ollama | ~7,300 LOC, non-clinical, synthetic-data-only

- Agentic RAG + Ingestion: built a citation-grounded document agent with automated ingestion from OpenFDA, PubMed, clinical-guideline, and media sources.
- RAFT Fine-Tuning: built a training-data pipeline with reference evidence, hard distractors, and abstention cases; fine-tuned Qwen2.5-1.5B via 4-bit QLoRA (rank 16, alpha 16), quantized it to GGUF, and served it locally via Ollama with human approval gates for data intake.

#### Other Projects

- TalentOS (All Things Agentic Hackathon, Taskmaster Track): dual-pipeline opportunity platform using Google ADK, LangGraph, and Firestore to collect ~2,900 postings/run, apply deterministic pre-filtering that eliminates 89.5% before any LLM call, and generate human-reviewable application materials, backed by a 373-test hermetic offline suite (~11s). FundersAI Reports: decoupled LangGraph report-synthesis service hosted on K3s. SQuAD QA Benchmarking: fine-tuned and benchmarked BERT, BiDAF, and DistilBERT; published as a research paper. GitResume: self-built evidence-backed resume CLI (PyPI: git-resume-agent, ~4K LOC, 28 commits) that validates managed claims before compiling Word/PDF output.

### EDUCATION & ACHIEVEMENTS

#### B.Tech, Computer Science Engineering (AI & ML Specialization) — VIT Amaravati

June 2021 – Sept 2025

CGPA: 8.2/10 | Published research paper on QA/NLP | Certifications: Azure Administrator Associate, AWS Certified Cloud Practitioner